

Answer all questions.

- 1 Solve the equation $\log_3(x + 4) - 4\log_9 3 = \log_9(x - 4)$. [6]

2 (a) (i) State the range of the principal values of $\tan^{-1} k$, where k is a real number. [1]

(ii) Without using a calculator, find, in radians, the principal value of $\tan^{-1} (\cos \pi)$. [2]

(b) Given that $\sin A = \frac{1}{2}$ and $\tan B = -\sqrt{3}$, where A and B are in the same quadrant, find the exact value of $\cos(A - B)$. [3]

3 Do not use a calculator in this question.

It is given that $x^2 - x - 6$ is a factor of $2x^4 + px^3 - 5x^2 + qx - 12$, where p and q are constants.

- (a)** Find the values of p and q . [6]

(b) Hence, factorise $2x^4 + px^3 - 5x^2 + qx - 12$ completely.

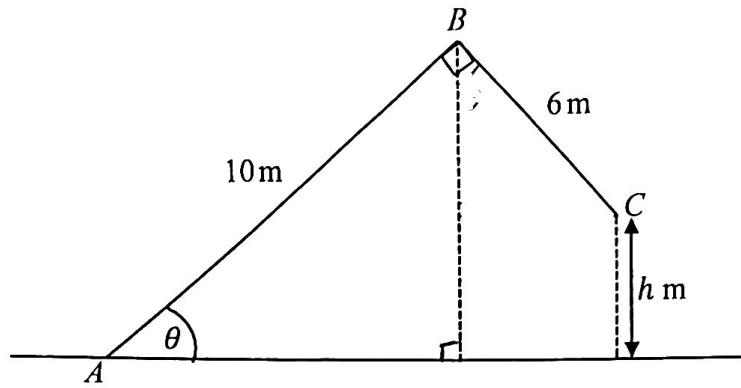
[3]

- 4 (a) Given that $\sqrt{6 - 4\sqrt{2}} = \sqrt{x} - \sqrt{y}$, find the value of x and of y . [5]

- (b) Simplify the expression $\frac{\sqrt{75} - \sqrt{72}}{\sqrt{48} - \sqrt{8}}$.

Leave your answer in the form $\frac{a+b\sqrt{6}}{20}$, where a and b are integers. [4]

5



The diagram shows two arms AB and BC of a construction crane rigidly joined at B so that angle $ABC = 90^\circ$. The lengths of AB and BC are 10 m and 6 m respectively. The point A is fixed on horizontal ground and the arm AB rotates in a vertical plane with the arm AB inclined at an angle θ to the ground, where $0^\circ < \theta < 90^\circ$.

- (a) Show that the height, h metres, of C above the ground is given by [2]

$$h = 10 \sin \theta - 6 \cos \theta.$$

- (b) Express h in the form $R \sin(\theta - \alpha)$, where $R > 0$ and α is acute. [4]

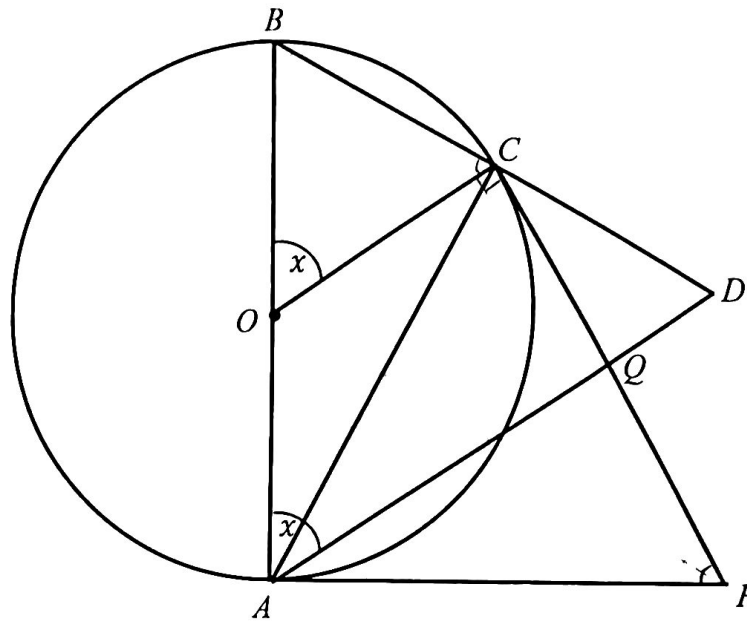
(c) Find the value of θ when C is 5 m above the ground.

[3]

(d) Find the value of θ when C touches the ground.

[2]

6



The diagram shows a circle with centre O . The points A , B and C lie on the circle. AB is a diameter, and BCD is a straight line. Angle $BOC = \text{Angle } BAD = x$. PA and PC are tangents to the circle at A and C respectively. AD and PC meet at Q .

(a) Prove that $OAPC$ is a kite.

[2]

(b) Prove that angle $BOC = \text{angle } APC$.

[4]

Continued space for (b)

(c) Explain why AC is the angle bisector of angle BAD .

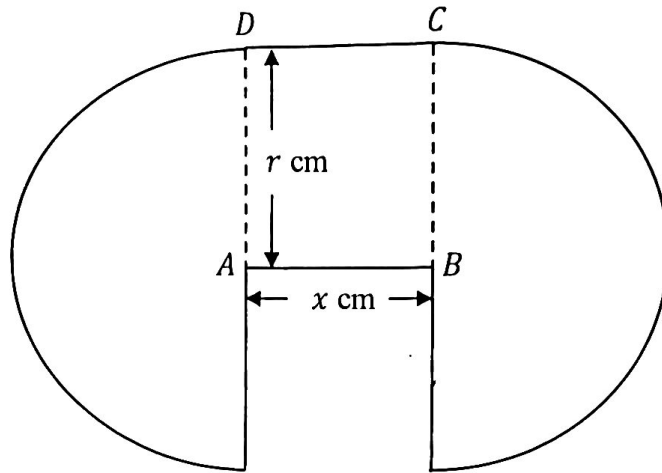
[1]

(d) Prove that C is the midpoint of BD .

(e) Prove that AB is a tangent of a circle which passes through points A , P and Q .

[1]

7



The diagram shows a badge, made of thin metal sheet, consisting of two semi-circular pieces, centres A and B , each of radius r cm. They are attached to a rectangular piece of thin metal sheet, $ABCD$, with $AB = CD = x$ cm.

The area of the badge is 30 cm^2 .

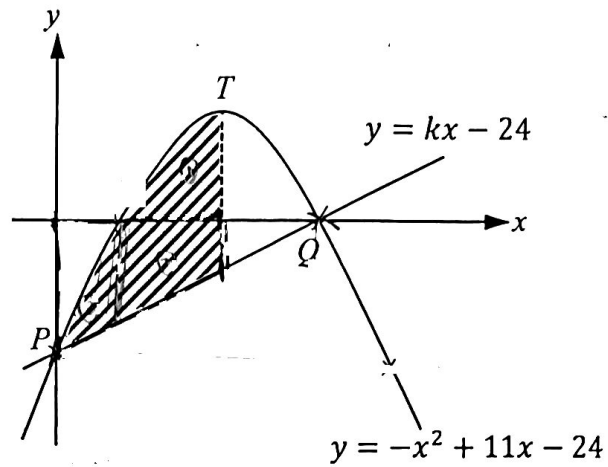
- (a) Show that the perimeter, P cm, of the badge is given by

$$P = 2r + \frac{60}{r}. \quad [3]$$

- (b) Given that r can vary, find the minimum value of P and justify that this value is a minimum. [5]

- (c) Prove that x is a decreasing function of r . [2]

8



The diagram shows the line $y = kx - 24$, where k is a constant, and the curve $y = -x^2 + 11x - 24$. The line intersects the curve at point P on the y -axis and point Q on the x -axis. Point T is the maximum point of the curve.

(a) Find the coordinates of T .

[3]

(b) Find the value of k .

[2]

- (c) Find the area of the shaded region enclosed by the curve, the line $y = kx - 24$, and the vertical line passing through T .

[5]

- 9 A particle P moves in a straight line.

The acceleration of P , $a \text{ ms}^{-2}$, t seconds after passing a fixed point, O , is given by

$$\begin{aligned} a &= 6 - 3e^{2-2t} & \text{for } 0 \leq t < 1, \\ a &= 4t^{-2} - t & \text{for } 1 \leq t \leq 5. \end{aligned}$$

When $t = 2$, P is instantaneously at rest and its displacement from O is $6\frac{2}{3} \text{ m}$.

- (a) Find the velocity of P when $t = 1$.

[3]

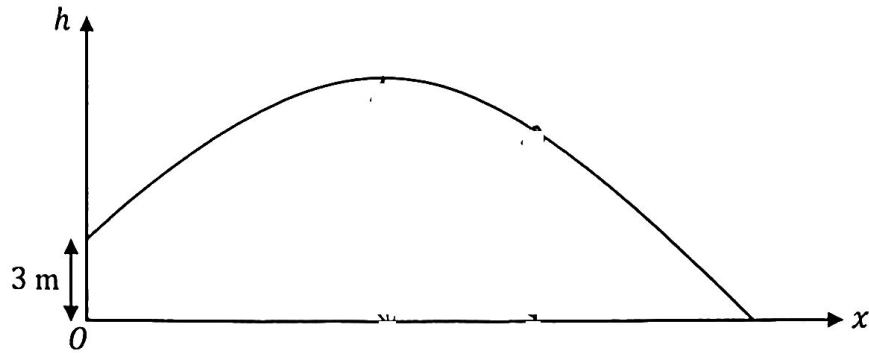
- (b) Find the displacement of P when $t = 1$.

[4]

(c) Find the initial velocity of P .

[4]

10



The diagram above shows the trajectory of a golf ball after it has been hit from a platform 3 metres above ground until it first touches the ground.

The height of the ball above the ground, h metres, has been plotted against the horizontal distance, x metres, from the bottom of the platform.

The ball reaches its maximum height after travelling a horizontal distance of 80 m.

The ball reaches a height of 24 m above the ground after travelling a horizontal distance of 100 m.

In this question, h is modelled as a **quadratic** function in x .

(a) Find h in terms of x .

[4]

- (b) The trajectory of an identical ball after it has been hit from a different platform is modelled by another equation

$$h = -0.05x^2 + 5x + 4.$$

A vertical pole of height 5 metres is newly installed on the ground.
This ball touches the top of the pole.

- (i) Give a physical interpretation of the meaning of the constant term 4 in this model. [1]

- (ii) Find the greatest horizontal distance of the pole from the platform. [3]

End of Paper